Lesson 4 Calculate ring-width indices on paper

For Exercise B on slide 24. About six minutes; no computer or CSV import is needed. These are synthetic values for practising the calculation, separate from the measured Nabunken data.

Each cell gives observed width / supplied growth-curve value, both in mm. Index = observed width divided by curve value. Curve fitting itself is not part of this exercise.

Year	Tree A	Tree B	Tree C
1998	2.26560 / 1.88800	1.92698 / 1.60582	1.09175 / 1.21306
1999	1.65469 / 1.83854	1.24045 / 1.55056	1.18460 / 1.18460
2000	1.96986 / 1.79078	1.94705 / 1.49773	—

Calculate each index to two decimal places. Average only available trees with equal weights, and record n. What happens if missing C in 2000 is replaced by zero?

Check

Year	Indices A, B, C	Mean	n
1998	1.20, 1.20, 0.90	1.10	3
1999	0.90, 0.80, 1.00	0.90	3
2000	1.10, 1.30, missing	1.20	2

Filling the missing value with zero gives a mean of 0.80, inventing a growth decline that was not observed. Do not count multiple radii from one tree as independent trees.

Return to the measured data

In the original 29ho018 CSV, 32 means 0.32 mm. The first seven widths sum to 2.79 mm. This is a sum of published measurements, not a new measurement from the printed photo. Check the original scale and measurement path; document positions and uncertainty if remeasuring.

29ho018: source image and measurements